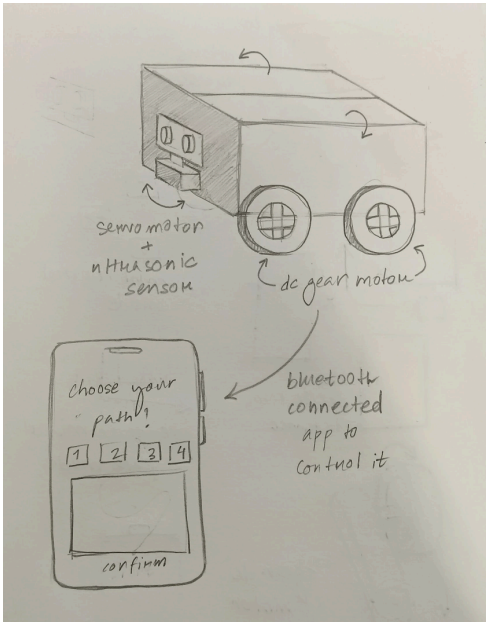
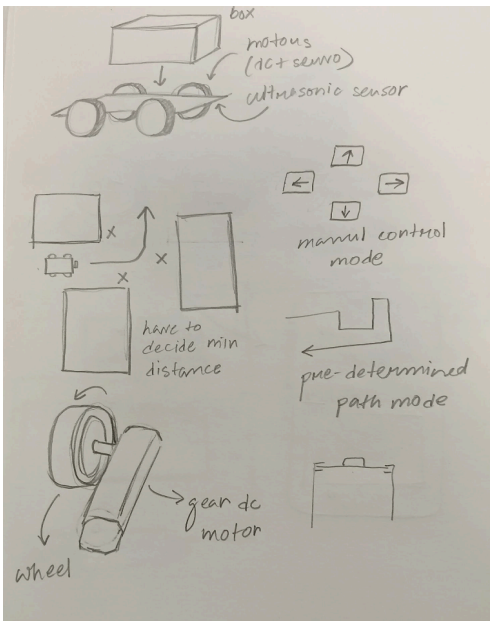
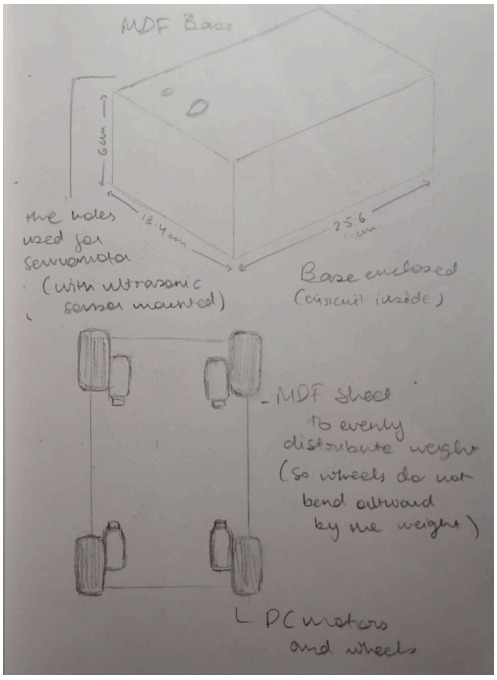
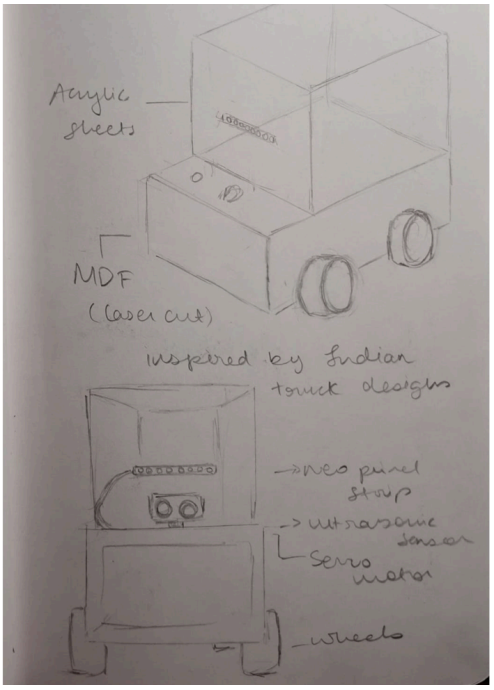
Week 1

1. We started by brainstorming and finalising the main idea for the ODT final assignment, making sure it was both practical and interesting to develop.
2. After that, we identified the key components required for the project, such as the ultrasonic sensor, DC motors, and servo motor, and clearly defined what role each of them would play in the overall system.
3. We then researched the specifications and requirements of these components and arranged for all the necessary parts to be sourced.
4. As a team, we discussed and planned out the logical flow of the code beforehand so that the development process would be more organised and efficient.
5. We created multiple rough sketches to visualise different design possibilities and gradually refined them to arrive at the most suitable form for the product.
6. Based on both functionality and appearance, We finalised the structure and material choices, selecting laser-cut MDF for the base due to its strength and stability, and laser-cut acrylic for the delivery box because of its clean and modern look.
7. Responsibilities were then divided among the team, with each member taking charge of specific coding sections and fabrication tasks to ensure smooth progress.
8. Finally, We designed a detailed circuit diagram to bring all the ideas together and clearly map out how each component would be connected and integrated into the system.


servo motor
+
ultrasonic
sensor
dc gear motor
choose your path!
1
2
3
4
confirm
bluetooth
connected
app to
control it


box
motors
(DC+SERVO)
ultrasonic sensor
manual control
mode
have to
decide min
distance
pre-determined
path mode
gear dc
motor
wheel


MDF Base
6cm
12.4cm
25.6
cm
the holes
used for
servomotor
(with ultrasonic
sensor mounted)
Base enclosed
(circuit inside)
MDF sheet
to evenly
distribute weight
(so wheels do not
bend outward
by the weight)
DC motors
and wheels


Acrylic
sheets
MDF
(laser cut)
inspired by Indian
truck designs
neo pixel
strip
ultrasonic
sensor
Servo
motor
wheels